ASSIGNMENT #1

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Course: Software Engineering - CSC112
Instructor: Mr. Muhammad Ghazi Abbas
Program: BS CS - IET - Spring 25 Morning
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Explain
Question 1: Software Engineering as Discipline & as a subject.

Solution:

Software Engineering as a Discipline

Software Engineering as a discipline that applies engineering concepts to design, build, test, and maintain a software. It's a structured approach to building software solutions, aiming for quality, efficiency and reliability. While often considered as a branch of computer science. It's distinct in its focus on a systematic and disciplined approach to software production. It's a field that requires specialised knowledge, skills, and tools to

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effectively develop and maintain software systems. So Software engineering is considered a discipline due to its systematic approach, emphasis on quality, and the use of engineering principles to solve complex programs.
In short, it's provides a framework for building reliable, efficient & high quality software solutions.

Key Aspects

1. Engineering Principles
It applies principles like modularity, abstractions and encapsulation, which're used in other engineering fields to create maintainable and scalable software.

2. Life Cycle Management
It consider the entire Software Development lifecycle from requirements gathering to deployement.

3. Tools & Techniques
Software engineers utilize a range of tools and techniques, including programming languages, software design tools and testing frameworks, to facilitate the development process.

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Software Engineering as a subject

Software Engineering as an academic subject equips students with theoritical knowledge and practical skills for software development. Undergraduate and graduate programs covers programming languages, system design, testing methodologies, data structures, algorithms and project management. The curriculum emphasizes problem solving, teamwork, and using advanced tools to build high quality software systems.

Question # 2.

Future of Software Engineering

The Future of software engineering is about to change a lot. Because of new technology and changing industry needs. Generally, software engineering combines ideas from computer science, engineering and project management to create applications and systems that help users and businesses. With more automation and Artificial Intelligence (AI) the way software is developed is changeing, bringing both new changes and challenges for workers in this feild by

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in upcoming few years but it doesn't means that it will be replaced by AI (Gen AI especially) but the way will be completely changed.

As we look at future of software engineering, it is clear that these advancements will shape how software is created and the skills needed for success in the industry. So A major trend shaping the future of SE is growing use of automation & AI. New tools that can handle coding, testing & launching software will be more smart and effective which allows SEngineers to spend less time on repetitive tasks and more time on important stuff like designing etc.

Question #3
Agile Process Model

Agile refers to something that is quick or adaptable. A software development approach based on iterative development is refered to as an "agile

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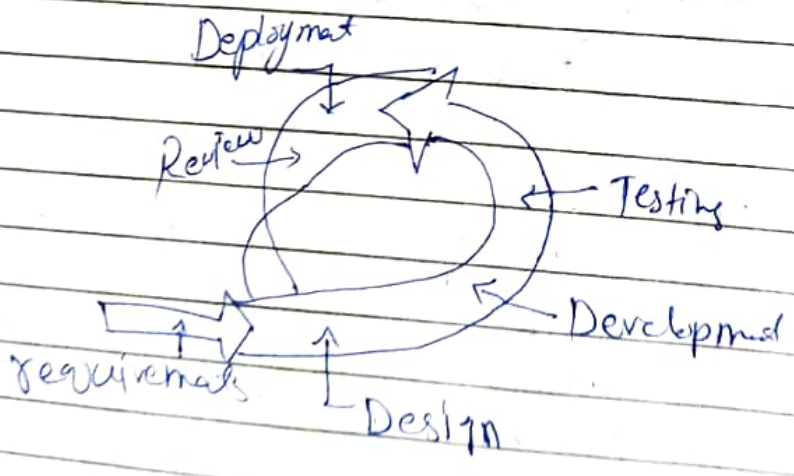
process model". Agile "Iterative" divides projects into smaller iterations or sections and avoid long-term planning. The scope and requirements of the project are defined at the start of development phase. The number of iterations are all clearly determined ahead of time. It is mostly used in large-scale projects.

Agile modeling is a strategy for demonstrating and documenting programming frameworks in light of best practices. It is a collection of values and standards that can be applied to programming development projects.



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When to use Agile model

- To execute a new feature, the developers need to lose just crafted by a couple of days or even just hours.
- At the point when new changes are required to have been carried out.
- End client's requirements are truly changing in a powerful business and IT world.

Pros

- Advances teamwork and broadly educating.
- Usefullness can be grown quickly and illustrated
- Asset requirements are minimum.

Cons

- Not appropriate for handling complex conditions.
- There's an absence of emphasis on vital planning & documentation.
- If client's requirements are not clear, project can be track off.

The End

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